

Flow-Based Video Recognition

Talk pipeline

- Introduction
- Sparse Feature Propagation
- Multi-frame Feature Aggregation
- Towards High Performance Video Object Detection (for Mobiles)
- Summary

From image to video



image semantic segmentation

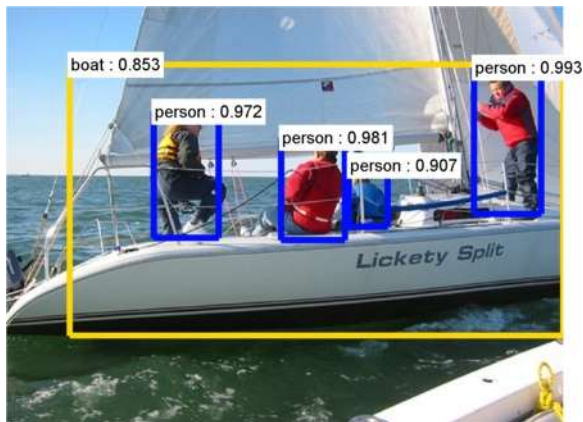
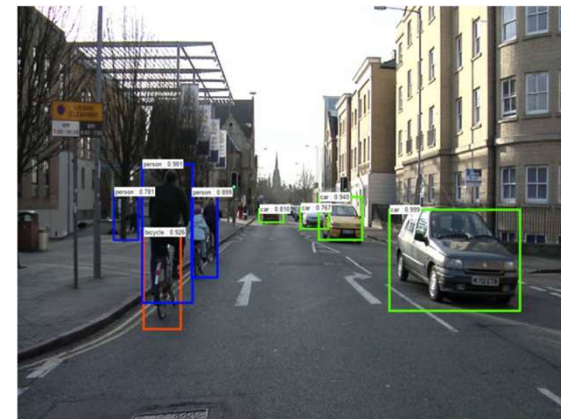


image object detection



video semantic segmentation



video object detection

Per-frame recognition in video is problematic

High Computational Cost

Infeasible for practical needs

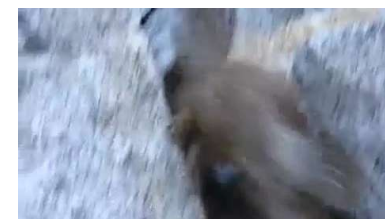
Task	Image Size	ResNet-50	ResNet-101
Detection	1000x600	6.27 fps	4.05 fps
Segmentation	2048x1024	2.24 fps	1.52 fps

FPS: frames per second
(NVIDIA K40 and Intel Core i7-4790)

Deteriorated Frame Appearance

Poor feature and recognition accuracy

motion
blur



part
occlusion



rare
poses



Exploit frame motion to do better (key idea)

- Sparse feature propagation (CVPR 2017)
 - Propagate features on sparse key frames to others
 - Up to **10x** faster at moderate accuracy loss
- Multi-frame feature aggregation (ICCV 2017)
 - Aggregate features on near-by frames to current frame
 - Enhanced feature, better recognition result
- Joint training of flow and recognition in DNN
- Clean, end-to-end, general
- Powering the winner of ImageNet VID 2017



key frame



flow field



current frame

Exploit frame motion to do better (application)

- Towards high performance video object detection (CVPR 2018)
 - Based on the key ideas
 - Accurate and fast
- Towards high performance video object detection on mobiles
 - The key ideas work at scarce computing resource
 - Accurate, real-time on mobiles for the first time
 - An order faster than the fastest object detectors with on par accuracy



key frame



flow field

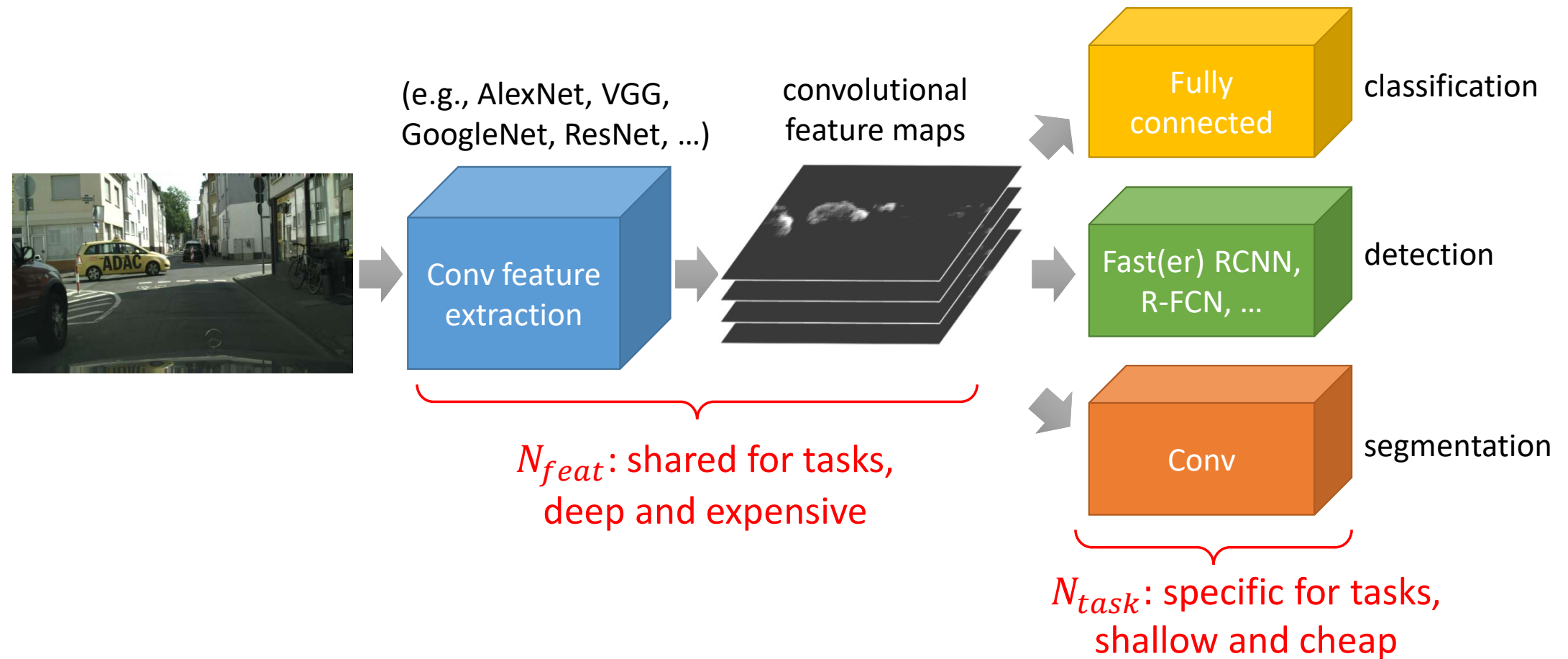


current frame

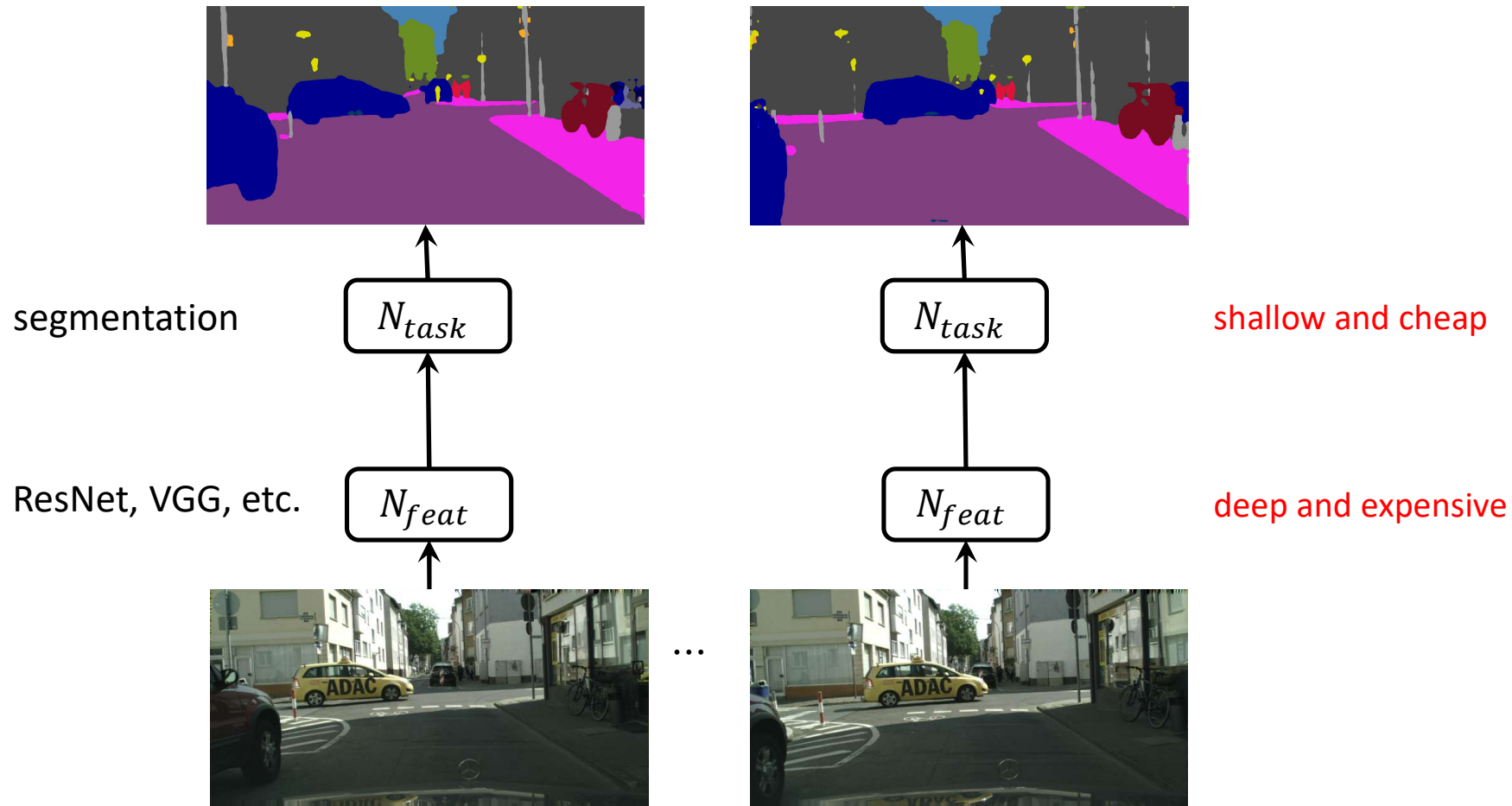
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Modern structure for image recognition



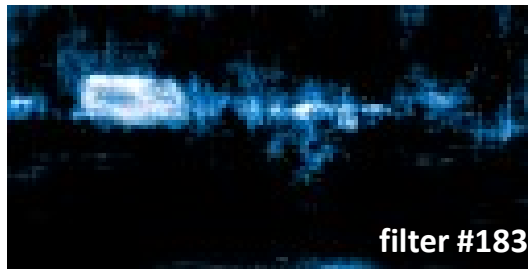
Per-frame baseline



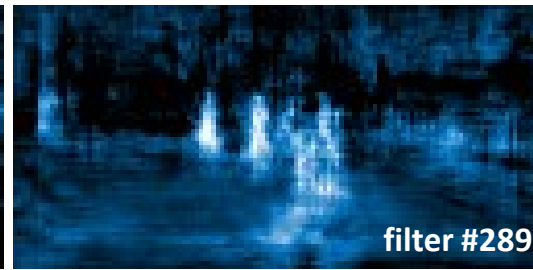
Sparse feature propagation: key idea



key frame



filter #183

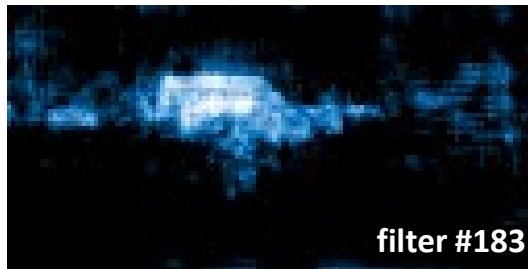


filter #289

key frame feature maps



current frame



filter #183

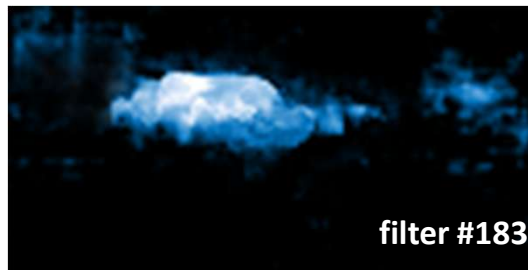


filter #289

current frame feature maps



flow field



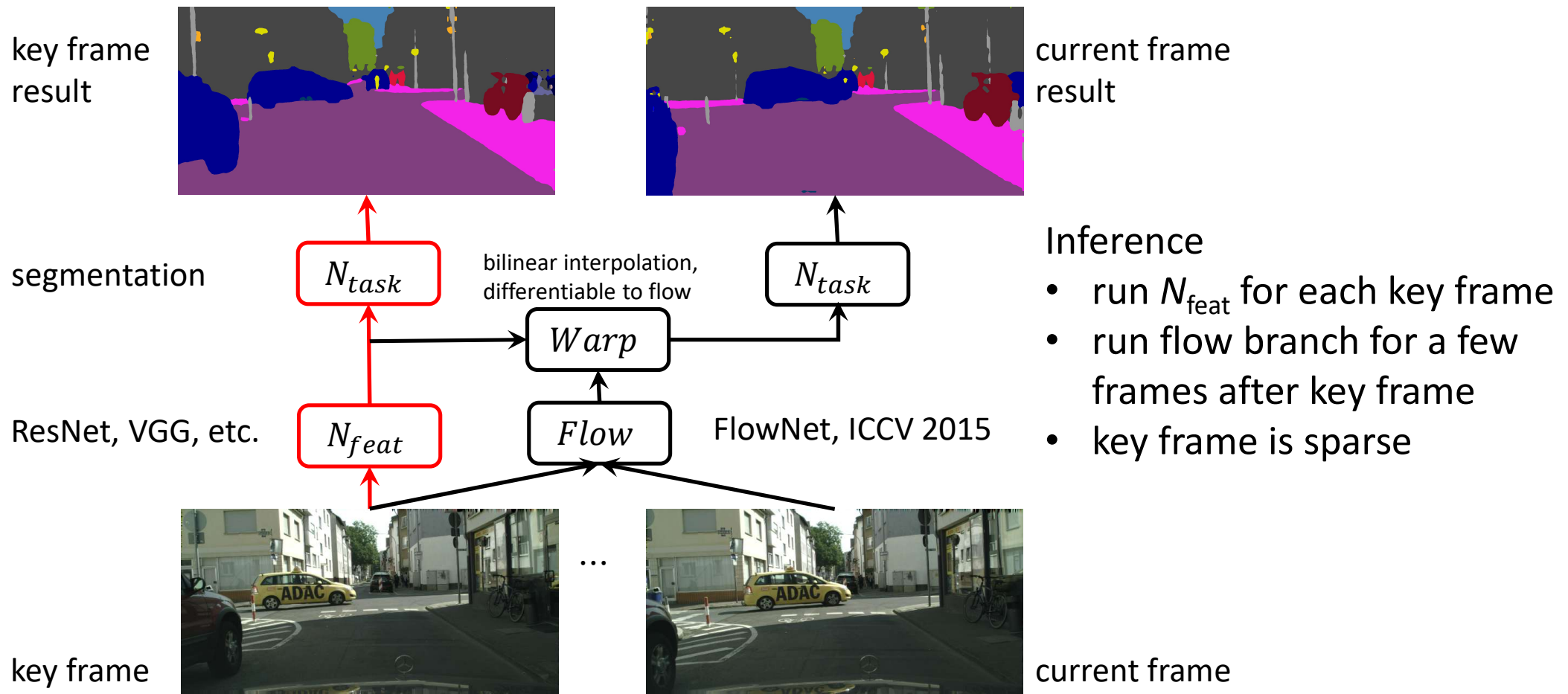
filter #183



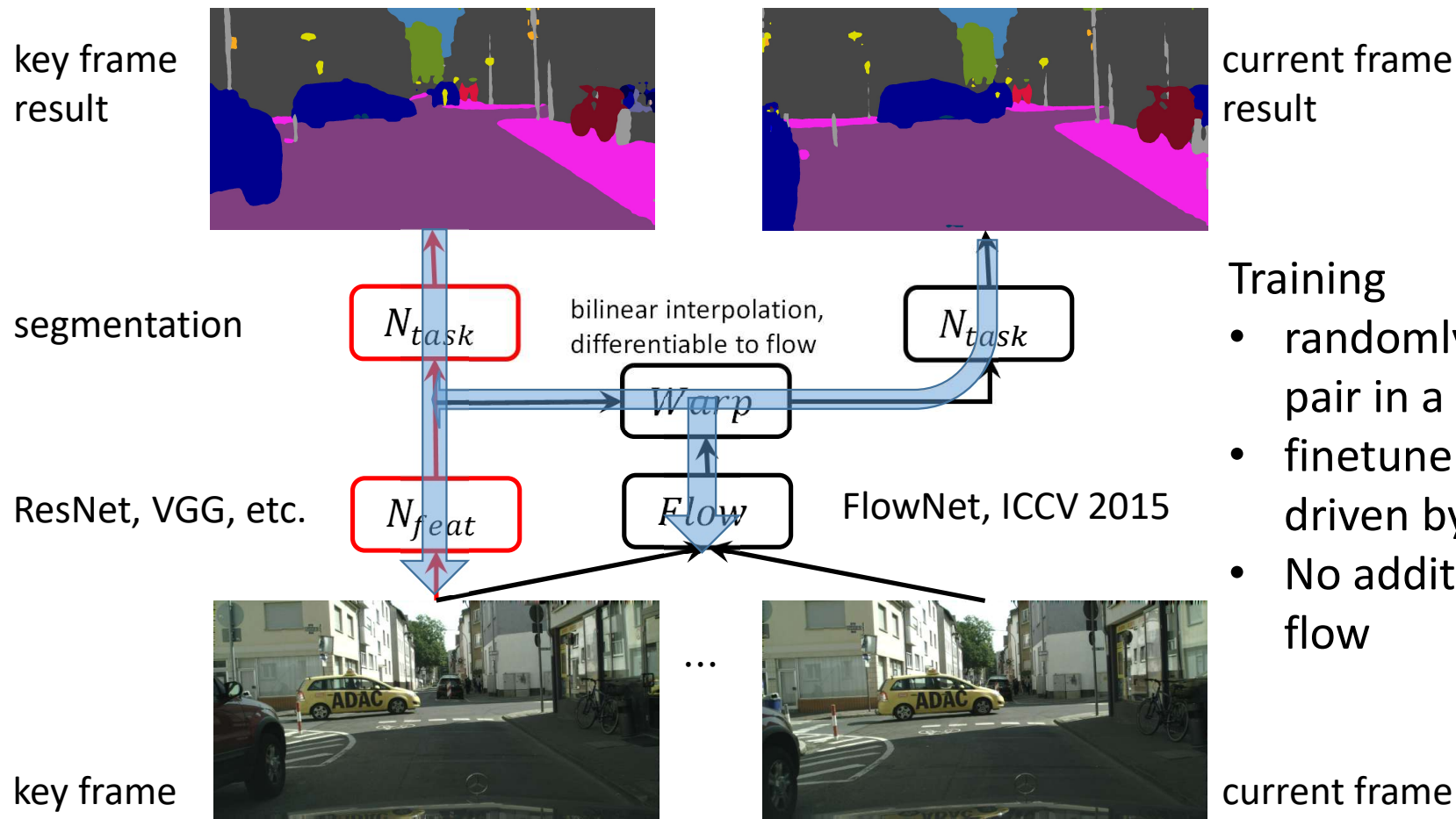
filter #289

warped from key frame to current frame

Sparse feature propagation: network structure



Sparse feature propagation: training



Training

- randomly sample a frame pair in a minibatch
- finetune all the modules driven by the recognition task
- No additional supervision for flow

Computational complexity analysis

- propagation from key frame W and N_{task} are very cheap

Per-frame computation ratio $r = \frac{O(F)+O(W)+O(N_{task})}{O(N_{feat})+O(N_{task})} \approx \frac{O(F)}{O(N_{feat})} \ll 1$

computation on key frame
- Flow F is much cheaper than feature extraction N_{feat}

$N_{feat} \setminus F$	FlowNet	FlowNet Half (1/4 of FlowNet)	FlowNet Inception (1/8 of FlowNet)
ResNet-50	9.20	33.56	68.97
ResNet-101	12.71	46.30	95.24

default setting

As $r \ll 1$, here we show $\frac{1}{r}$ for clarify.

Experiment datasets

task	semantic segmentation	object detection
dataset	CityScapes	ImageNet VID
frames per second	17	25 or 30
key frame duration	5	10
#semantic categories	30	30
#videos	train 2975, validation 500, test 1525	train 3862, validation 555, test 937
#frames per video	30	6~5492
annotation	every 20 th frame	all frames
evaluation metric	mIoU (mean of Intersection-over-Union)	mAP (mean of Average Precision)

key frame duration is manually chosen to fit the application needs for accuracy-speed trade-off

1. a long duration saves more feature computation but has lower accuracy as flow is less accurate
2. vice versa for a short duration

Ablation study: results on two tasks

method \ task	segmentation	on CityScapes	detection	on ImageNet VID
method \ metric	mIoU (%)	runtime (fps)	mAP (%)	runtime (fps)
<i>Frame</i> (oracle baseline)	<u>71.1</u>	1.52	<u>73.9</u>	4.05
<i>SFF</i> : shallow feature flow (SIFT)				
<i>SFF-slow</i>	67.8	0.08	70.7	0.26
<i>SFF-fast</i>	67.3	0.95	69.7	3.04
<i>DFF</i> : deep feature flow				
<i>DFF</i>	69.2	5.60	73.1	20.25
<i>DFF fix N</i>	68.8	5.60	72.3	20.25
<i>DFF fix F</i>	67.0	5.60	68.8	20.25
<i>DFF separate</i>	66.9	5.60	67.4	20.25

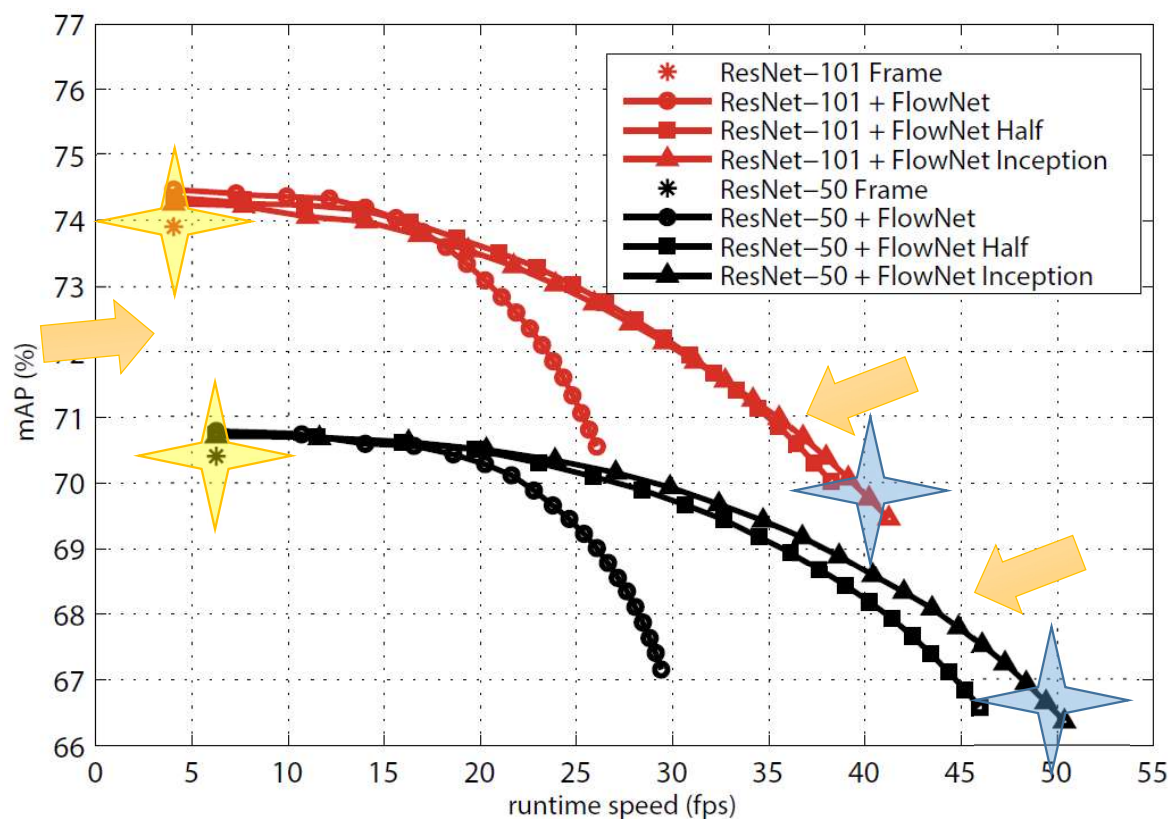
1. ***DFF*** is much faster than single *Frame* baseline at moderate accuracy loss

2. Using off-the-shelf flow algorithm is worse

3. Joint end-to-end training is effective

Accuracy-speedup tradeoff by varying N_{feat} and F

- Significant speedup with decent accuracy drop
(10X faster, 4.4% accuracy drop)
- How to choose flow function?
 - Cheapest FlowNet Inception is the best
- How to choose conv. features?
 - ResNet101 is better



ImageNet VID detection (5354 videos, 25 ~ 30 fps)

Cityscapes Dataset (17 fps, 1024 x 2048)

only **single** frame is **annotated** in each snippets (30 frames)



Ground truth



Our results

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Deteriorated appearance in videos

motion
blur



video
defocus



part
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rare
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How to improve video object detection

Post-processing: box level

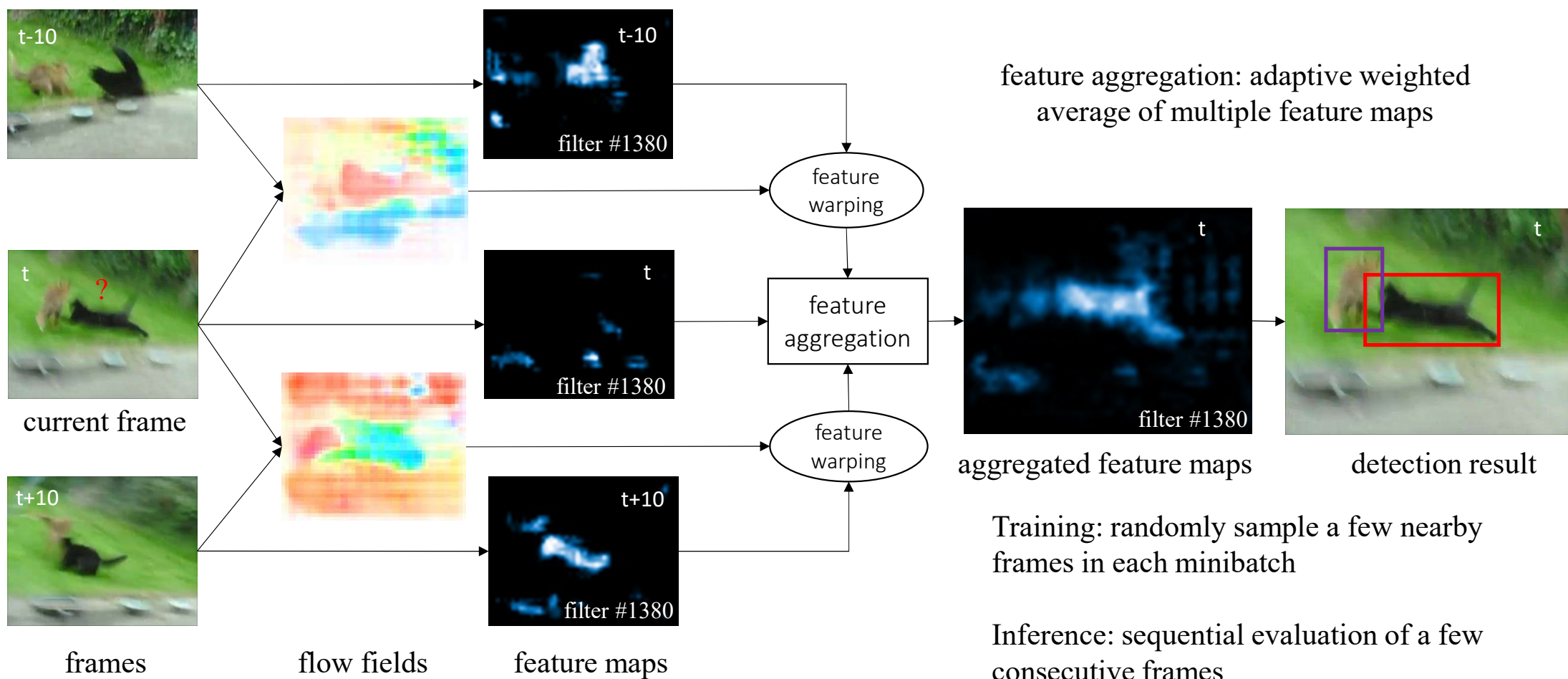
- Manipulation of detected boxes
 - e.g., **tracking** over multi-frames
- Heuristic, heavily engineered
- Widely used in competition

Better feature learning: feature level

- Enhance deep features
 - **learning** over multi-frames
- Principled, clean
- Rarely studied

First end-to-end DNN work for video object detection

Flow-guided feature aggregation

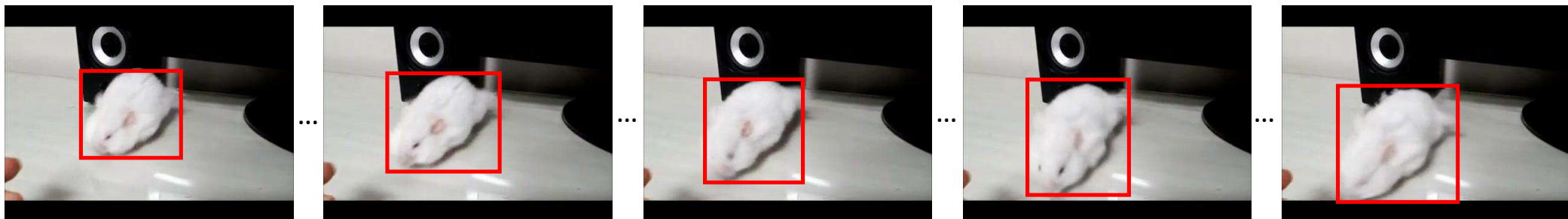


Use motion IoU to measure object speed

slow



medium



fast



t-10

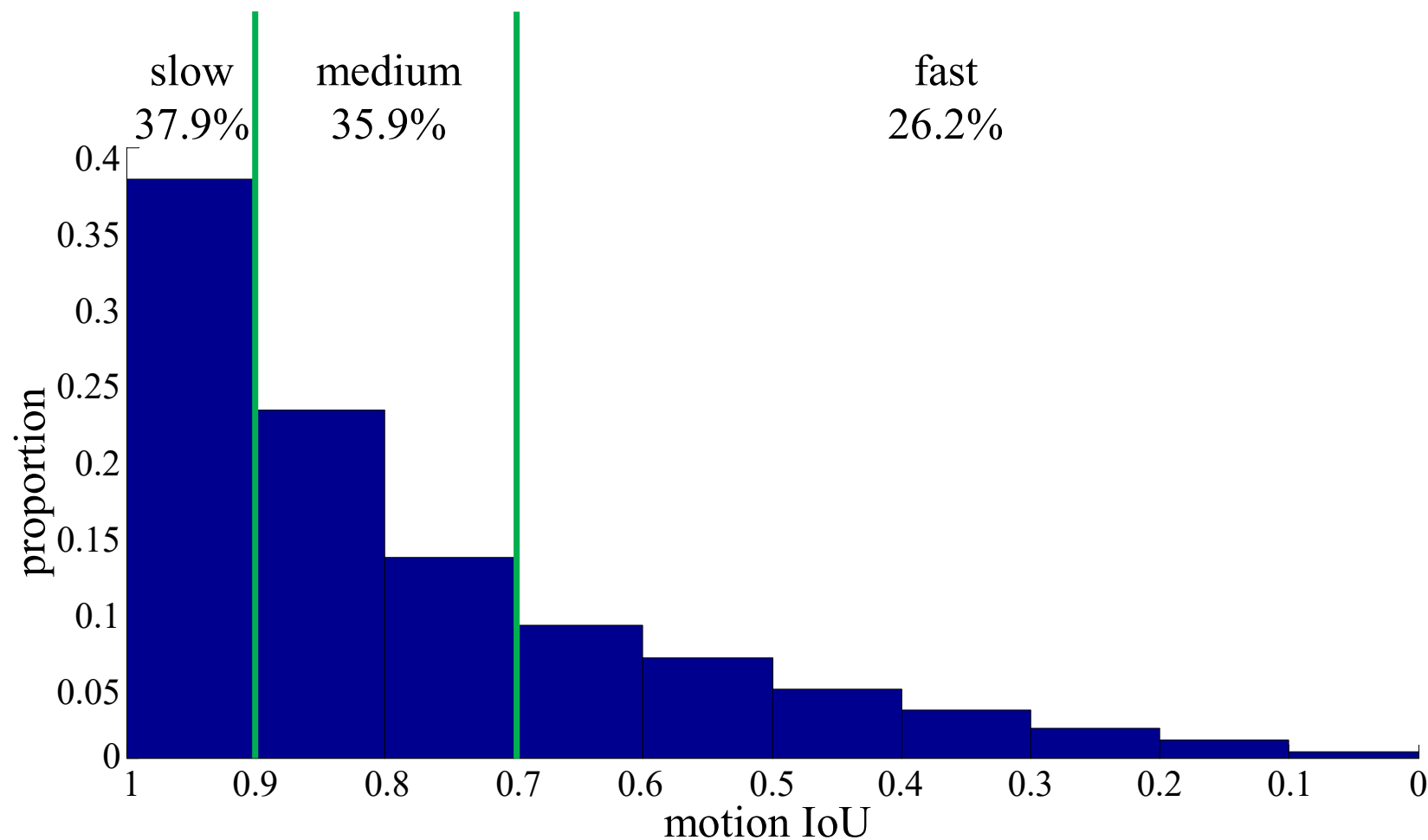
t-5

t

t+5

t+10

Categorization of object speed



Ablation study results on ImageNet VID

methods	Single frame baseline	Ours (no flow/weights)	Ours (no flow)	<i>Ours</i>	Ours (no e2e training)
multi-frame aggregation		✓	✓	✓	✓
adaptive weights			✓	✓	✓
flow guided				✓	✓
end-to-end training		✓	✓	✓	
mAP (%)	73.4	72.0	74.3	76.3 (↑2.9)	74.5
mAP (%) (slow)	82.4	82.3	82.2	83.5 (↑1.1)	82.5
mAP (%) (medium)	71.6	74.5	74.6	75.8 (↑4.2)	74.6
mAP (%) (fast)	51.4	44.6	52.3	57.6 (↑6.2)	53.2
runtime (ms)	288	288	305	733	733

1. All components (flow, adaptive weighting, end-to-end learning) are important.
2. Especially effective on fast (difficult) objects
3. Slower as flow computation takes time

#frames in training and inference

#test frames	1	5	9	13	17	21*	25
mAP (%) 2* frames in train	70.6	72.3	72.8	73.4	73.7	74.0	74.1
mAP (%) 5 frames in train	70.6	72.4	72.9	73.3	73.6	74.1	74.1
runtime (ms)	203	330	406	488	571	647	726

*: default parameter

- More frames in inference is better (saturated at 21)
- 2 frames in training is sufficient (frame skip randomly sampled)

Integration with post-processing techniques

- Complementary with post-processing techniques
- A clean solution with state-of-the-art performance (80.1 mAP)
 - ImageNet VID 2016 winner: 81.2
 - Highly engineered with various tricks, not used in ours

method	feature network	mAP (%)	runtime (ms)
single-frame baseline	ResNet-101	73.4	288
+ MGP		74.1	574*
+ Tubelet rescoring		75.1	1662
+ Seq-NMS		76.8	433*
FGFA	ResNet-101	76.3	733
+ MGP		75.5	1019*
+ Tubelet rescoring		76.6	1891
+ Seq-NMS		78.4	873*
FGFA	Aligned-Inception-ResNet	77.8	819
+ Seq-NMS		80.1	954*

Table 4. Results of baseline method and FGFA before and after combination with box level techniques. As for runtime, entry marked with * utilizes CPU implementation of box-level techniques.

Powering the winner of ImageNet VID 2017

Team name	Entry description	Number of object categories won	mean AP
IC&USYD	provide_submission3	15	0.817265
IC&USYD	provide_submission1	6	0.808847
IC&USYD	provide_submission2	4	0.818309
NUS-Qihoo-UIUC_DPNs (VID)	no_extra + seq + mca + mcs	3	0.757772
NUS-Qihoo-UIUC_DPNs (VID)	no_extra + seq + vcm + mcs	1	0.757853
NUS-Qihoo-UIUC_DPNs (VID)	Faster RCNN + Video Context	1	0.748493
THU-CAS	merge-new	0	0.730498
THU-CAS	old-new	0	0.728707
THU-CAS	new-new	0	0.691423
GoerVision	Deformable R-FCN single model+ResNet101	0	0.669631
GoerVision	Ensemble 2 model, use ResNet101 as fundamental classification network and deformable R-FCN to detect video frames, multi-scale testing	0	0.665693
GoerVision	o train the video objectWe use the ResNet101 and Deformable R-FCN for the detection.	0	0.655686
GoerVision	Using R-FCN to detect video object, multi scale testing applied.	0	0.646965
FACEALL_BUPT	SSD based on Resnet101 networks	0	0.195754

[\[top\]](#)

IC&USYD	Jiankang Deng(1), Yuxiang Zhou(1), Baosheng Yu(2), Zhe Chen(2), Stefanos Zafeiriou(1), Dacheng Tao(2), (1)Imperial College London, (2)University of Sydney Flow acceleration[1,2] is used. Final scores are adaptively chosen between the detector and tracker. [1] Deep Feature Flow for Video Recognition Xizhou Zhu, Yuwen Xiong, Jifeng Dai, Lu Yuan, and Yichen Wei, IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017. [2] Flow-Guided Feature Aggregation for Video Object Detection, Xizhou Zhu, Yujie Wang, Jifeng Dai, Lu Yuan, and Yichen Wei. Arxiv tech report, 2017.
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Video demo

Results

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Revisiting Two Baseline Methods on Video

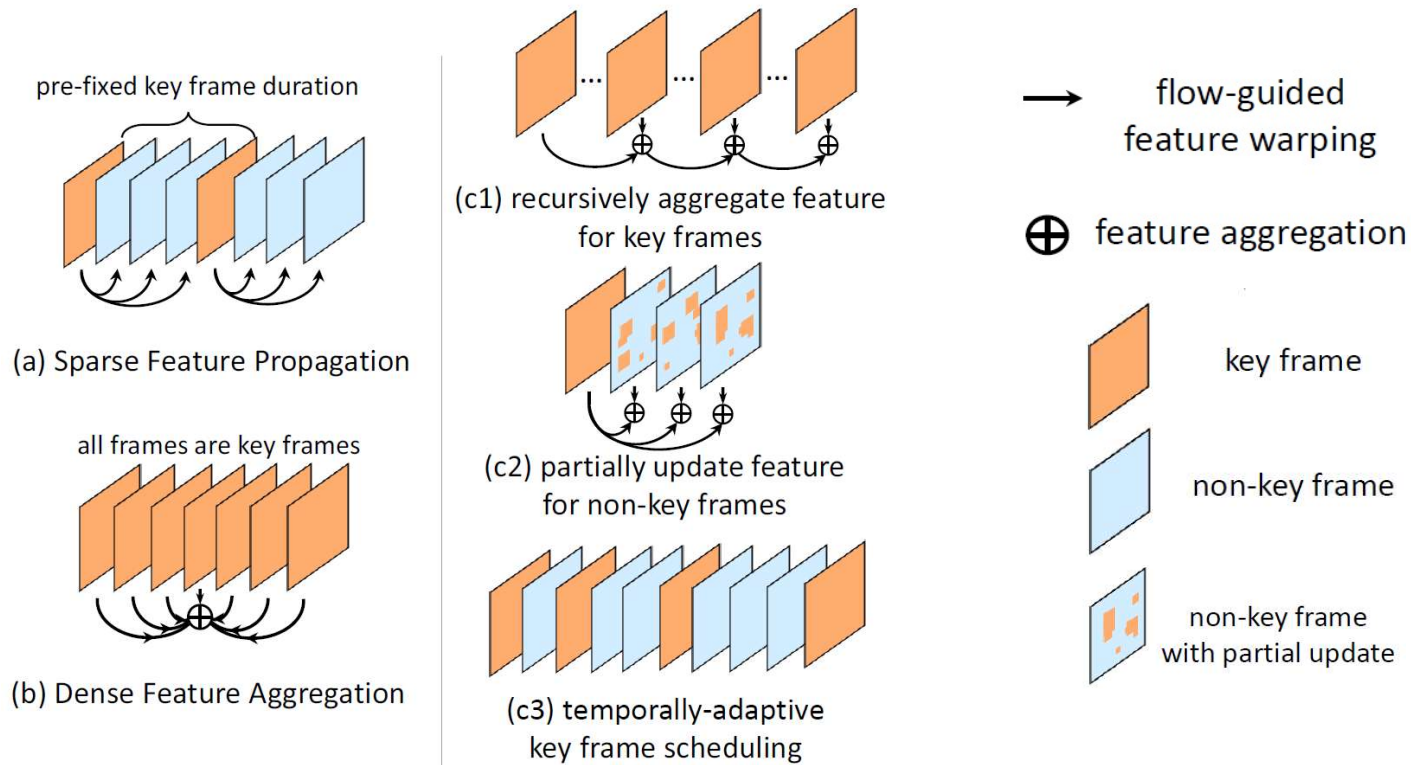
- Sparse Feature Propagation
 - Introduce the concept of key frame
 - Reduce feature computation by feature approximation, decreased accuracy
- Multi-frame Feature Aggregation
 - Introduce the concept of temporal feature aggregation
 - improve feature quality by adaptive aggregation, increased computation
- They are naturally complementary

Towards High Performance Video Object Detection

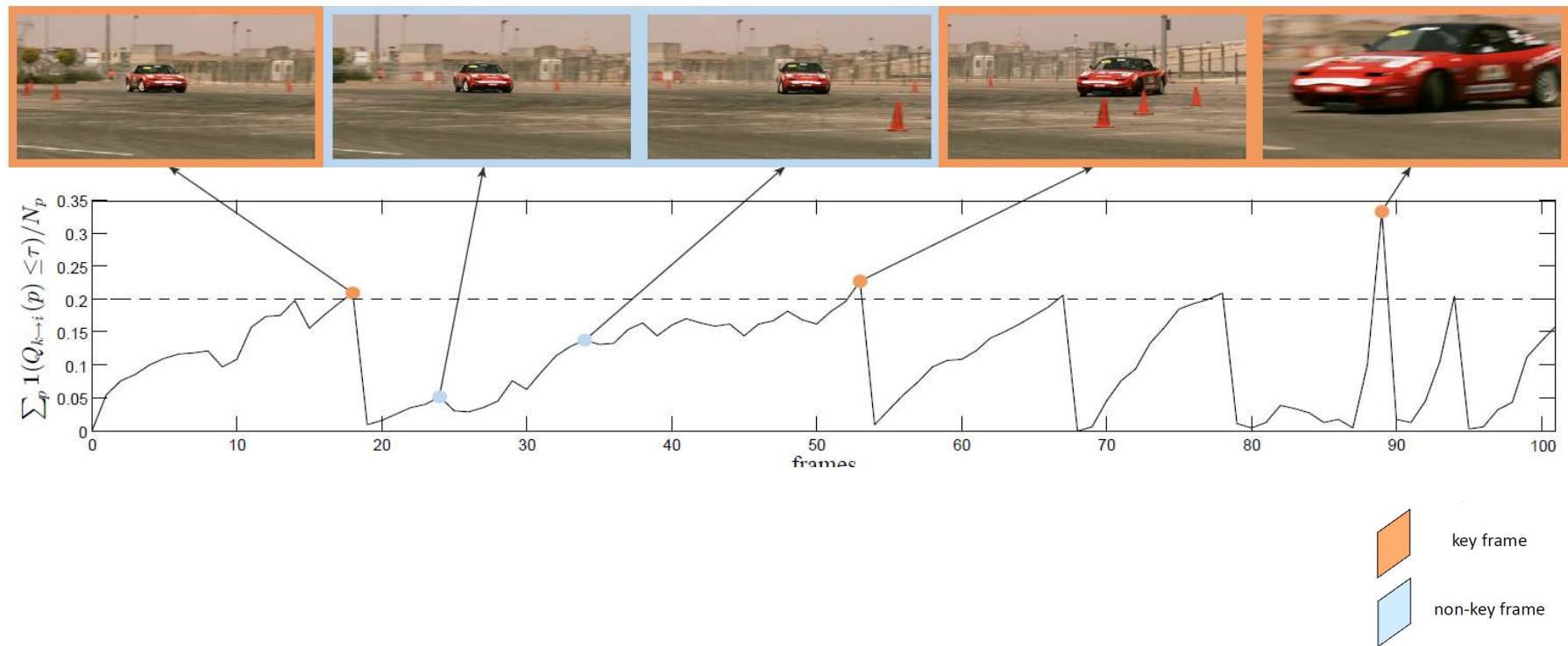
- Based on two principles (as verified in the two baselines)
 - *motion estimation* module is indispensable for effective feature level communication between frames
 - *end-to-end learning over multiple frames* of all modules is crucial for detection accuracy

Towards High Performance Video Object Detection

- Extend prior works with three new techniques

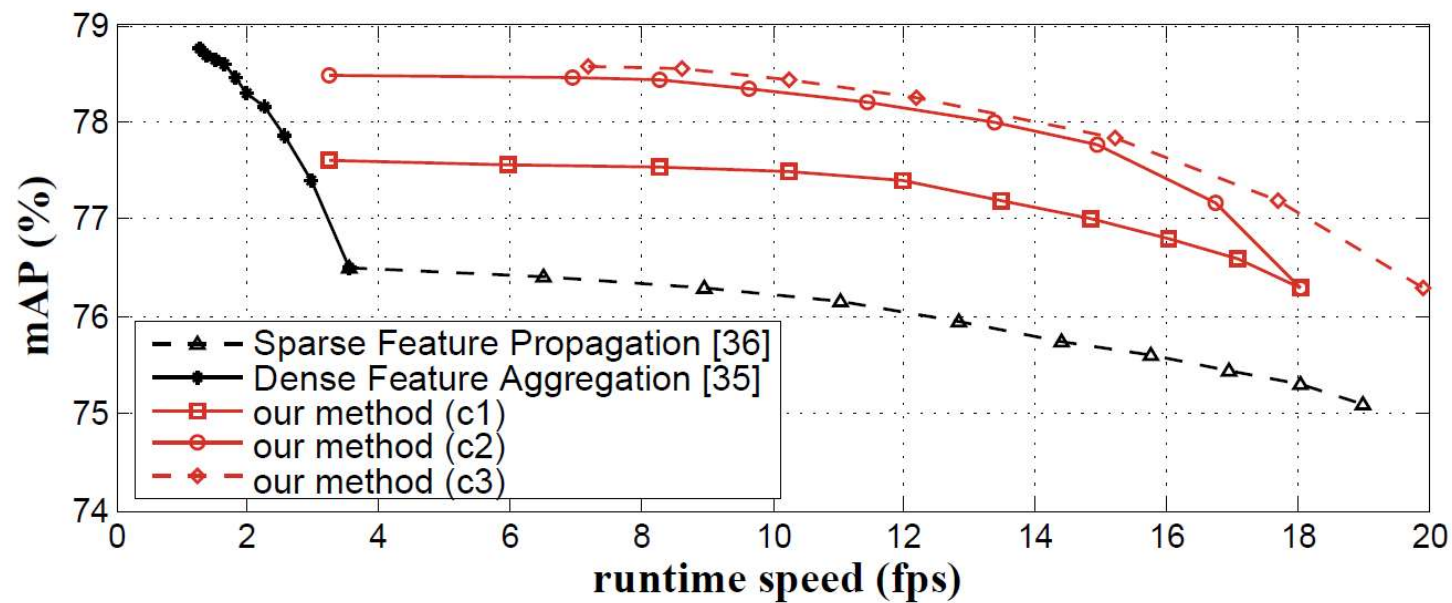


Towards High Performance Video Object Detection



Experiments

- Push forward the speed-accuracy trade-off envelope

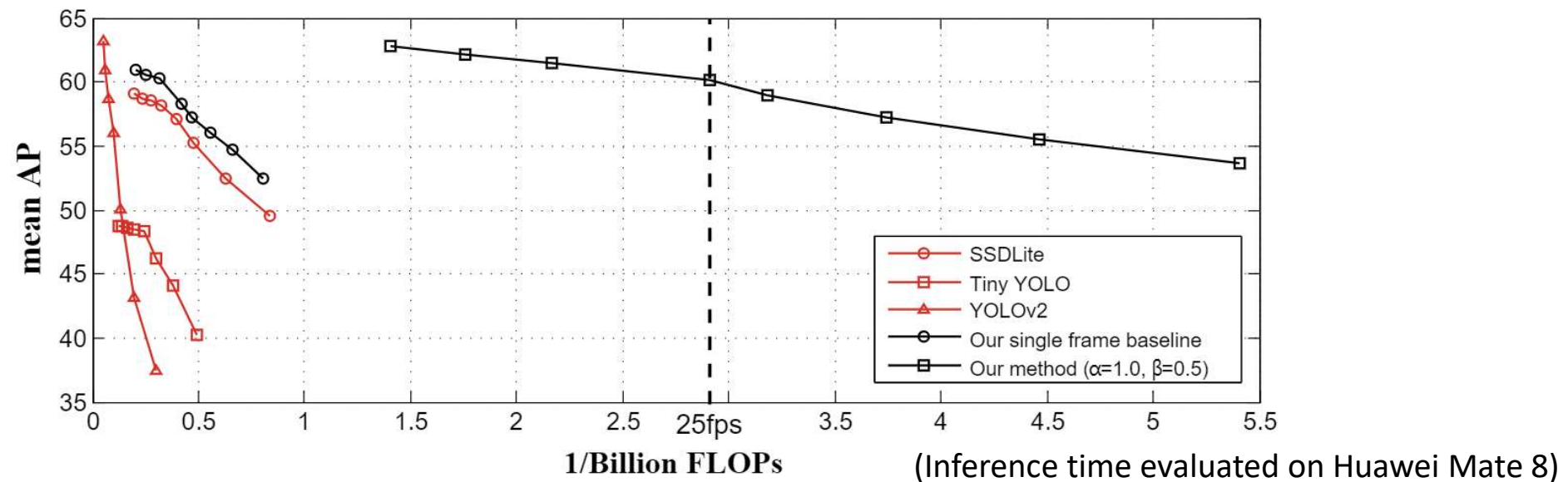


Challenges of Video Object Detection for Mobiles

- Too heavy network architectures for mobiles
 - ResNet & FlowNet are too heavy for mobiles
 - Lack of ultra light flow network
- Unclear application of the key principles at scarce computing resource
 - Are sparse feature propagation and multi-frame feature aggregation still beneficial?

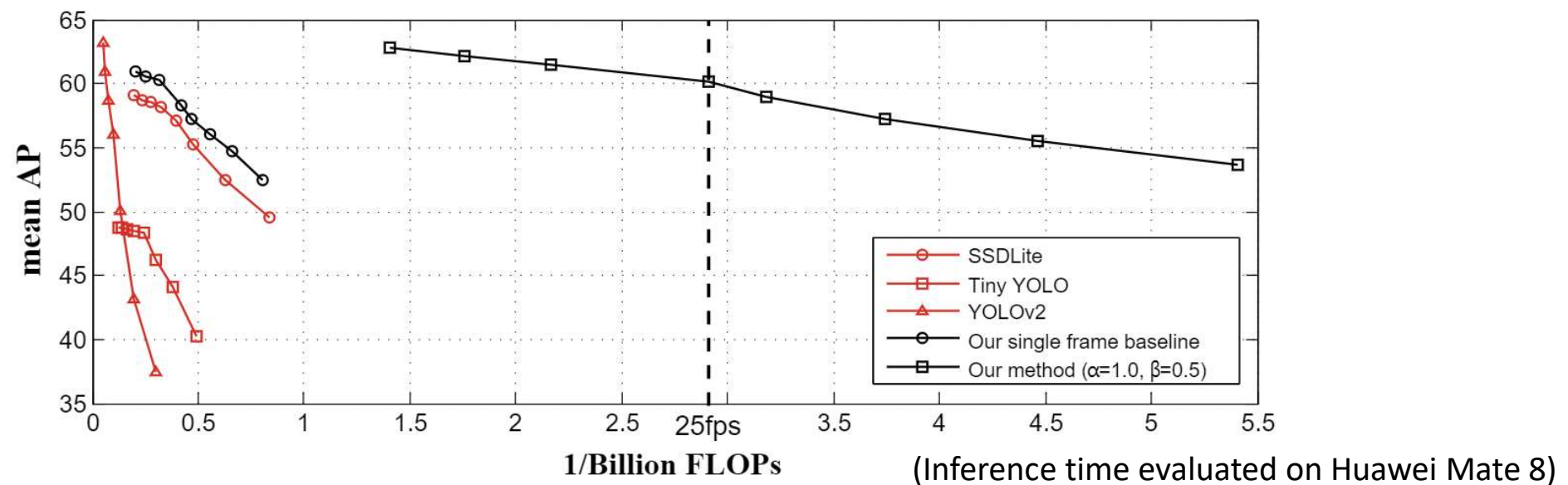
Towards High Performance Video Object Detection for Mobiles (Under Review)

- Single frame baseline: MobileNet + Light-head R-CNN
- Correspondence establish: design of an ultra light network, Light Flow
- The principles still hold at scarce computing resource!



Towards High Performance Video Object Detection for Mobiles (Under Review)

- Accurate, real-time video object detection on mobiles for the first time
- An order faster than previous fastest object detectors with on par accuracy



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Summary

- Exploit motion for video recognition tasks
 - Faster speed AND better accuracy
- End-to-end, joint learning of optical flow and recognition tasks
- Feature learning instead of heuristics, general for different tasks
- Code available at
 - <https://github.com/msracver/Deep-Feature-Flow>
 - <https://github.com/msracver/Flow-Guided-Feature-Aggregation>

Future work

- Better flow learning and evaluation
- Better key frame scheduling
 - Better efficiency and accuracy, simultaneously
- Low latency on mobile devices
- Joint learning for detection and tracking
 - new losses (smoothness, box association) on temporal dimension
 - On the stability of video detection and tracking, arXiv 2016

Thanks! Q & A